



VR2Tx FATHOM DATA FORMAT

Fathom format v2.0 specification

DOC-8638-01

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1. Overview

This document describes the data records and fields available for export to CSV from VR2Tx receiver log files (.VRL or .VDAT), using Innovasea’s Fathom tools.

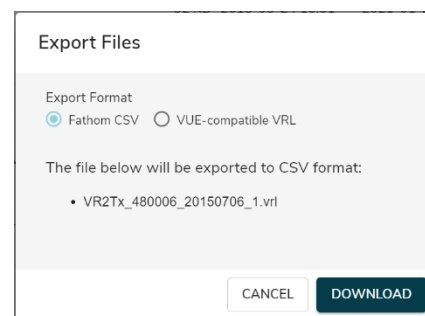
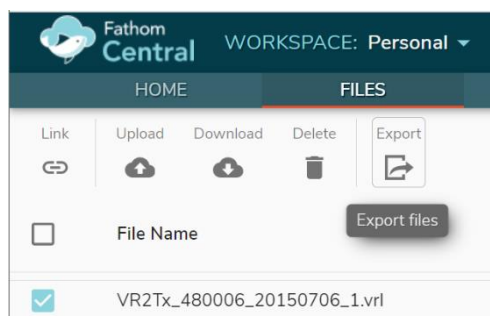
In this document, the following terms are used:

- Data Record: a row of data containing columns (data fields) related to a particular data point. For example, a **detection** data record.
- Data Field: a particular column of a data record. For example, a detection **time**.
- Data Type: a group of data records that are related to each other. For example, **Configuration** records.

Receiver logs can be exported to CSV using either the Fathom Central web application or the Fathom Connect desktop application. Refer to the sections below for details on each export method. To skip to the data record and data field definitions for the VR2Tx, refer to **Section 3, Data Field Descriptions**.

1.1 Receiver Log Export Method 1: Fathom Central

1. Visit <https://fathomcentral.com/> and signup or login to your account
2. If your receiver logs were offloaded using VUE (.VRL) or Fathom Connect (.VDAT), upload your files to your preferred Fathom Central workspace.
3. If your receiver logs were offloaded using the Fathom Mobile app (.VDAT) and you have synced them to Fathom Central, proceed to the workspace you had selected in the Mobile app to locate your files.
4. In the “Files” tab in Fathom Central, select the files you would like to export to CSV and click “Export”
5. Choose “Fathom CSV” in the resulting “Export Files” dialog.



After exporting your Fathom CSV file(s) using this method, please proceed to **Section 2, How to Use Fathom CSV Files**.

1.2 Receiver Log Export Method 2: Fathom Connect

1. Download and install Fathom Connect



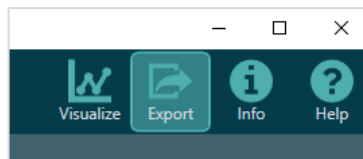
**Fathom
Connect**



Scan or visit

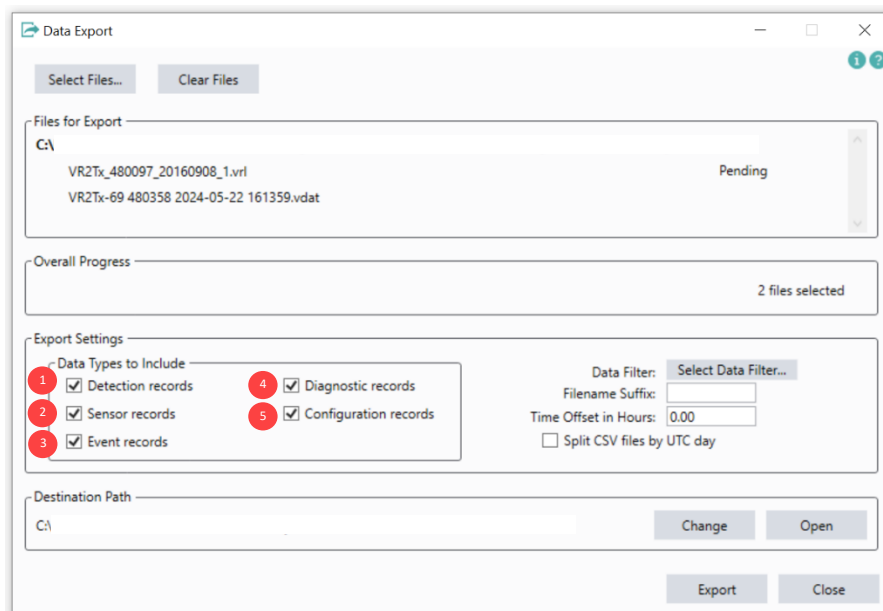
<https://support.fishtracking.innovasea.com/s/downloads> to
download the Fathom Connect software

2. In Fathom Connect, select “Export”.



3. Choose the desired data types (defined below) to export. For information on the other export options in the Fathom Connect Data Export window, please refer to the Fathom Connect User Manual at the link provided in Step 1.

1.2.1 Fathom Connect Data Types



	Export Settings Option	Data Types Exported	Record Identifier in CSV File
1	Detection Records	Detection information	DET
2	Sensor Records ¹	Receiver temperature data	TEMP
		Receiver tilt data	ATTITUDE
3	Event Records	General receiver event	EVENT
		Receiver fault information	EVENT_FAULT
		Receiver initialization information	EVENT_INIT
		Receiver offload information	EVENT_OFFLOAD
4	Diagnostic Records	Battery and battery status information	BATTERY
		Diagnostic (system performance) data	DIAG ²
		Fast diagnostics (system performance) data	DIAG_FAST ³
		Receiver memory remaining and other health information	HEALTH_VR2TX
5	Configuration Records	Receiver channel configuration information	CFG_CHANNEL
		Receiver station information	CFG_STATION
		Study information	CFG_STUDY
		Receiver self-transmitter configuration information	CFG_TRANSMITTER
Exported with all Data Types		Information on external clock used to set receiver clock	CLOCK_REF
		Data error information	DATA_ERROR
		Original VDAT or VRL receiver file information	DATA_SOURCE_FILE

¹ VR2Tx receiver logs from firmware versions prior to 5.0.1 may also have TEMP_STATS records

² VR2Tx receiver logs from firmware versions prior to 5.0.1 will have DIAG_VR2TX and DIAG_VR2TX_INTERIM records instead

³ VR2Tx receiver logs from firmware versions 5.0.1 and later only

2. How to Use Fathom CSV Files

All Fathom CSV files consist of collections of multiple data records. Each row in the CSV file is either a record **Descriptor Row** (*_DESC) or a **Data Row**. Descriptor rows contain all of the column headings for the data in that record. Data rows contain the record's data.

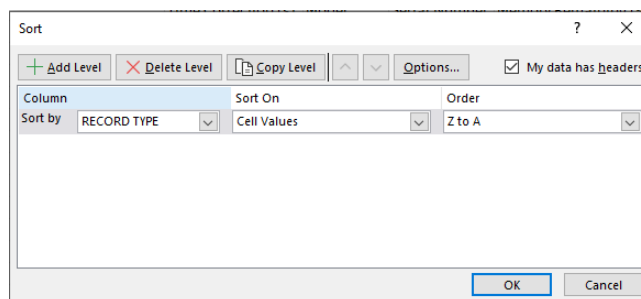
Below are some example descriptor rows (identified by the *_DESC suffix):

	A	B	C	D	E	F	G	H	I	J
2	RECORD TYPE	FIELD	FIELD	FIELD	FIELD	FIELD	FIELD	FIELD	FIELD	FIELD
3	ATTITUDE_DESC	Device Time (UTC)	Time	Time Offset (h)	Time Correction (s)	Model	Serial Number	Tilt (deg)		
4	BATTERY_DESC	Device Time (UTC)	Time	Time Offset (h)	Time Correction (s)	Model	Serial Number	Battery Position	Battery Type	Battery Serial Number
5	CFG_CHANNEL_DESC	Device Time (UTC)	Time	Time Offset (h)	Time Correction (s)	Model	Serial Number	Channel	Detection Type	Frequency (kHz)
6	CFG_STATION_DESC	Device Time (UTC)	Time	Time Offset (h)	Time Correction (s)	Model	Serial Number	Station Name	Latitude (deg)	Longitude (deg)
7	CFG_STUDY_DESC	Device Time (UTC)	Time	Time Offset (h)	Time Correction (s)	Model	Serial Number	Description		

Below are some example data rows (rows **without** *_DESC suffix):

12	CLOCK_REF	2023-11-15 13:53:53	2023-11-15 13:53:53		0 VR2Tx-69	499999	2023-11-15 13:53:53	0 INITIALIZATION		
13	CFG_CHANNEL	2023-11-15 13:53:53	2023-11-15 13:53:53		0 VR2Tx-69	499999		1 PPM	69	Generation-2
14	CFG_STUDY	2023-11-15 13:53:53	2023-11-15 13:53:53		0 VR2Tx-69	499999	Sturgeon Study			
15	CFG_TRANSMITTER	2023-11-15 13:54:11	2023-11-15 13:54:11		0.000395 VR2Tx-69	499999	PPM	A69-1601-99999	99999 HIGH	540 660
16	CLOCK_REF	2024-01-04 17:42:49	2024-01-04 17:42:49		0 VR2Tx-69	499999	2024-01-04 17:44:24	95 OFFLOAD		

To match the record descriptor rows (column headings) with the corresponding data for the record, sort the CSV file on the first ("RECORD TYPE") column, in descending order:



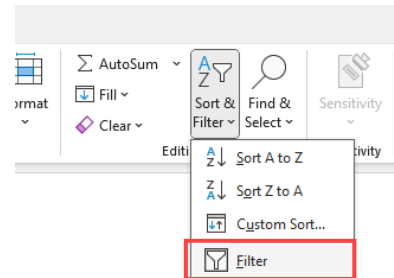
The record's column headings (in bold below for visibility) will now appear above the data for that record, as shown below:

	A	B	C	E	F	G	H	I	J
2	EVENT_OFFLOAD_DESC	Device Time (UTC)	Time	Time Correction (s)	Model	Serial Number	External Time (UTC)	External Time Zone	
3	EVENT_OFFLOAD	2024-01-04 17:42:49	2024-01-04 17:42:49	0	VR2Tx-69	499999	2024-01-04 17:44:24	UTC-06:00	
4	EVENT_INIT_DESC	Device Time (UTC)	Time	Time Correction (s)	Model	Serial Number	Prior Device Time (UTC)	External Time Zone	Firmware Version
5	EVENT_INIT	2023-11-15 13:53:53	2023-11-15 13:53:53	0	VR2Tx-69	499999	2023-11-15 13:54:03	UTC-06:00	7
6	EVENT_DESC	Device Time (UTC)	Time	Time Correction (s)	Model	Serial Number	Type	Subtype	Description
7	EVENT	2023-11-19 19:56:32	2023-11-19 19:56:40	8.052891315	VR2Tx-69	499999	TRANSPONDING_SESSION_BEGIN		
8	EVENT	2023-11-19 20:06:32	2023-11-19 20:06:40	8.066043939	VR2Tx-69	499999	TRANSPONDING_SESSION_END		
9	DET_DESC	Device Time (UTC)	Time	Time Correction (s)	Model	Serial Number	Channel	Detection Type	Full ID
10	DET	2023-11-15 13:54:16	2023-11-15 13:54:16	0.000504521	VR2Tx-69	499999		1 PPM	A69-1601-99999
11	DET	2023-11-15 13:55:02	2023-11-15 13:55:02	0.001514765	VR2Tx-69	499999		1 PPM	A69-1601-99999
12	DET	2023-11-15 13:55:22	2023-11-15 13:55:22	0.001949216	VR2Tx-69	499999		1 PPM	A69-1601-99999
13	DET	2023-11-15 13:55:44	2023-11-15 13:55:44	0.002431251	VR2Tx-69	499999		1 PPM	A69-1601-99999
14	CLOCK_REF_DESC	Device Time (UTC)	Time	Time Correction (s)	Model	Serial Number	External Time (UTC)	External Difference (s)	Source
15	CLOCK_REF	2023-11-15 13:53:53	2023-11-15 13:53:53	0	VR2Tx-69	499999	2023-11-15 13:53:53		0 INITIALIZATION
16	CLOCK_REF	2024-01-04 17:42:49	2024-01-04 17:42:49	0	VR2Tx-69	499999	2024-01-04 17:44:24	95	OFFLOAD

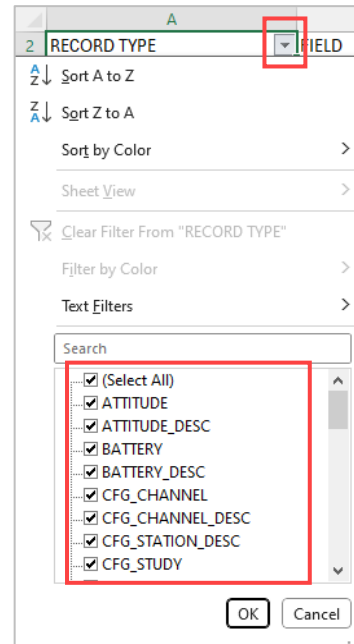
To view a specific data record of interest (for example, detection records only), use Excel's "Filter" feature as shown below. If you are using another program to manipulate your data (such as R), select your data based on the descriptors in the first column of the CSV file (for example, select **DET_DESC** and **DET** for detection records).

To filter your data in Excel:

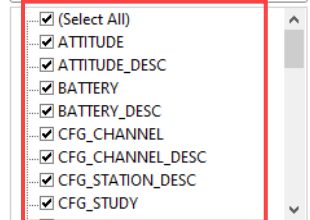
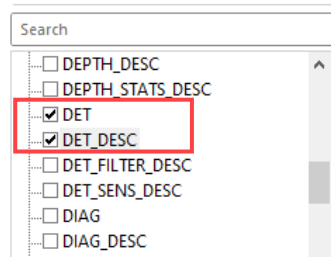
1. Select the entire worksheet and go to "Sort & Filter" in the Editing group in the top ribbon. Click "Filter".



2. Click on the down arrow next to RECORD TYPE in column A of the worksheet to see the list of all available records types in your CSV file.



3. Unselect all, then select the descriptor row and the data row for the record of interest only. Click "OK":



4. Your data are now filtered to show only the data record of interest.

	A	B	C	D	E	F	G	H	I	J
2	RECORD TYPE	FIELD	FIELD	FIELD	FIELD	FIELD	FIELD	FIELD	FIELD	FIELD
15	DET_DESC	Device Time (UTC)	Time	Time Offset (h)	Time Correction (s)	Model	Serial Number	Channel	Detection Type	Full ID
33	DET	2023-11-15 13:54:16	2023-11-15 13:54:16		0.000504521	VR2Tx-69	499999	1	PPM	A69-1601-99999
34	DET	2023-11-15 13:55:02	2023-11-15 13:55:02		0.001514765	VR2Tx-69	499999	1	PPM	A69-1601-99999
35	DET	2023-11-15 13:55:22	2023-11-15 13:55:22		0.001949216	VR2Tx-69	499999	1	PPM	A69-1601-99999
36	DET	2023-11-15 13:55:44	2023-11-15 13:55:44		0.002431251	VR2Tx-69	499999	1	PPM	A69-1601-99999
37	DET	2023-11-15 13:55:51	2023-11-15 13:55:51		0.002584376	VR2Tx-69	499999	1	PPM	A69-1601-99999

3. Data Field Descriptions

This section describes the data records and defines the data fields (columns) in each record.

3.1 Detection Records

DET records include tag detection information. Some fields in this record only apply to certain receiver models and tag types; these are indicated in the table where appropriate.

3.1.1 DET

Column	Field	Description
1	Record Descriptor	Either DET_DESC (row containing column names for the DET record) or DET (a detection data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Channel	Receiver channel number. Always Channel 1 for the VR2Tx.
9	Detection Type	Reports whether PPM or HR signal detected. Always PPM for the VR2Tx.
10	Full ID	Full ID of the detected transmitter including code space (ex: A69-1601-2056)
11	ID	The ID of the detected transmitter without the code space (ex: 2056)
12	Raw Data	A raw sensor value that is converted into engineering units (for example, °C or m) using the specifications of the sensor transmitter
13	Transmitter Serial	Unused for the VR2Tx
14	Signal Strength (dB)	
15	Noise (dB)	
16	Gain (dB)	

17	Quality Score	
18	Station Name	The name of the station where the receiver was deployed (if assigned)
19	Latitude	Receiver latitude (if known)
20	Longitude	Receiver longitude (if known)
21	GPS HDOP	Unused for the VR2Tx

3.2 Sensor Records

Sensor records contain data from the receiver's built-in sensors, logged at intervals chosen by the user.

3.2.1 ATTITUDE

ATTITUDE records contain receiver orientation (tilt) data from a receiver's built-in sensor. They are generated according to the logging interval selected in the Sensor Logging tab of the Fathom Connect software. Attitude (tilt) data are also available in the diagnostic (DIAG) record.

Column	Field	Description
1	Record Descriptor	Either ATTITUDE_DESC (row containing column names for the ATTITUDE record) or ATTITUDE (a tilt data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Tilt (deg)	The tilt of the receiver, in degrees from vertical

3.2.2 TEMP

Column	Field	Description
1	Record Descriptor	Either TEMP_DESC (row containing column names for the TEMP record) or TEMP (a temperature data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Ambient (deg C)	Ambient external temperature (°C)
9	Internal (deg C)	Unused for VR2Tx

3.3 Event Records

EVENT records track the occurrence of significant receiver events. All events except initialization, offload, and fault events are catalogued in the EVENT record. For initialization, offload, and receiver fault information refer to the EVENT_INIT, EVENT_OFFLOAD, and EVENT_FAULT records respectively.

3.3.1 EVENT

Column	Field	Description
1	Record Descriptor	Either EVENT_DESC (row containing column names for the EVENT record) or EVENT (an event data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Type	Event descriptor (ex: STARTUP, SHUTDOWN, TRANSPONDING_SESSION_BEGIN, TRANSPONDING_SESSION_END)
9	Subtype	Used for troubleshooting purposes. Only defined for SHUTDOWN record.
10	Description	Event description (where applicable)
11-17	Values	Series of values reported based on the event type (where applicable)
18	Latitude	Unused for VR2Tx
19	Longitude	
20	GPS HDOP	

3.3.2 EVENT_FAULT

EVENT_FAULT records track the occurrence of receiver faults for troubleshooting purposes.

Column	Field	Description
1	Record Descriptor	Either EVENT_FAULT_DESC (row containing column names for the EVENT_FAULT record) or EVENT_FAULT (an fault event data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Fault Importance	An indicator of the severity of the fault. Fault levels are as follows: LOW - No action required MEDIUM – Replace receiver battery (if indicated) or contact Innovasea Support HIGH – Discontinue receiver use and contact Innovasea Support immediately
9	Fault Code	A code used by Innovasea Support.
10	Fault Description	A brief description of the fault.

3.3.3 EVENT_INIT

EVENT_INIT records contain receiver initialization information.

Column	Field	Description
1	Record Descriptor	Either EVENT_INIT_DESC (row containing column names for the EVENT_INIT record) or EVENT_INIT (an initialization event data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock. This time is the same as the UTC time of the device (PC, mobile device) that was used to initialize the receiver.
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column. NOTE: for an EVENT_INIT record, the time-corrected time will be the same as the uncorrected time.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times. NOTE: For EVENT_INIT records, time correction will be 0.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Prior Device Time (UTC)	UTC date and time of the receiver's clock prior to being set according to the external device clock at initialization
9	External Time Zone	Time zone of the device that was used to set the receiver's clock at initialization
10	Firmware Version	Firmware version of the receiver at initialization

3.3.4 EVENT_OFFLOAD

EVENT_OFFLOAD records contain receiver offload information. Some fields in this record only apply to certain receiver models; these are indicated in the table where appropriate.

Column	Field	Description
1	Record Descriptor	Either EVENT_OFFLOAD_DESC (row containing column names for the EVENT_OFFLOAD record) or EVENT_OFFLOAD (an offload event data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	External Time (UTC)	UTC date and time of the device that was used to offload the receiver (calculated from the device's clock time and time zone)
9	External Time Zone	Time zone of the device that was used to offload the receiver
10	HR Total Accepted Detections	Unused for VR2Tx
11	PPM Total Accepted Detections	Count of all detections on the receiver at offload
12	Memory Remaining (%)	Percentage of total memory capacity remaining at offload
13	Battery Remaining (%)	Estimated percentage of battery life remaining at offload
14	Original File	The name that Fathom assigned to the offloaded receiver log file

3.4 Diagnostic Records

3.4.1 BATTERY

BATTERY records contain battery status information. They contain different information depending on the receiver model and the event that generated the record. All receiver models log a battery record at least once per day at midnight.

Column	Field	Description
1	Record Descriptor	Either BATTERY_DESC (row containing column names for the BATTERY record) or BATTERY (a battery data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Battery Position	Unused for VR2Tx
9	Battery Type	
10	Battery Serial Number	
11	Battery Voltage (V)	Battery voltage remaining
12	Battery Remaining (%)	Estimated percentage of battery life remaining

3.4.2 DIAG

DIAG records contain VR2Tx diagnostic information collected at hourly intervals. Ping and detection counts are calculated every hour, and temperature and tilt are sampled and logged hourly. Noise is sampled once per minute and averaged and logged hourly. This record applies only to receiver firmware versions 5.0.1 and later. For earlier firmware versions please refer to the DIAG_VR2Tx record.

Column	Field	Description
1	Record Descriptor	Either DIAG_DESC (row containing column names for the DIAG record) or DIAG (a diagnostic data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Ambient Temperature (deg C)	Ambient external temperature (°C), sampled at time of logging
9	Noise Mean (mV)	Hourly average noise (average of 60 samples at 1 sample / minute)
10	Tilt (deg)	Receiver tilt angle (degrees from vertical), sampled at time of logging
11	Depth (m)	Unused for VR2Tx
12	PPM Pings	Number of pings recorded during the previous hour
13	PPM Detections	Number of detections recorded during the previous hour

3.4.3 DIAG_FAST

DIAG_FAST records contain VR2Tx diagnostic information collected once per minute. Temperature, noise, and tilt are sampled and logged every minute for a period of 14 days following the activation of Fast Diagnostic Mode in the Fathom Connect software. This record applies only to receiver firmware versions 5.0.1 and later. For earlier firmware versions please refer to the DIAG_VR2Tx_INTERIM record.

Column	Field	Description
1	Record Descriptor	Either DIAG_FAST_DESC (row containing column names for the DIAG_FAST record) or DIAG_FAST (a fast diagnostics data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Ambient temperature (deg c)	Ambient external temperature (°C), sampled at time of logging
9	Noise (mV)	Ambient noise, sampled at time of logging
10	Tilt (deg)	Receiver tilt angle (degrees from vertical), sampled at time of logging
11	Depth (m)	Unused for VR2Tx

3.4.4 DIAG_VR2Tx (FW < v5.0.1 only)

DIAG_VR2Tx records contain VR2Tx diagnostic information collected at regular intervals. Ping and detection counts are calculated every 24 hours at midnight (UTC), and Noise and Tilt readings are sampled according to the interval selected in VUE's Logging Options window. Applies only to VR2Tx firmware versions prior to 5.0.1 and VUE versions prior to 2.6.0.

Column	Field	Description
1	Record Descriptor	Either DIAG_VR2TX_DESC (row containing column names for the DIAG_VR2TX record) or DIAG_VR2TX (a VR2Tx diagnostics data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Pings	Number of pings recorded in the previous 24 hours
9	Detections	Number of detections recorded in the previous 24 hours
10	Noise (mV)	Ambient noise, sampled at time of logging
11	Tilt (deg)	Receiver tilt angle (degrees from vertical), sampled at time of logging

3.4.5 DIAG_VR2Tx_INTERIM (FW < v5.0.1 only)

DIAG_VR2Tx_INTERIM records contain VR2Tx ping and detection count information collected at irregular intervals. Entries are typically associated with receiver events (such as offload, for example). Applies only to VR2Tx firmware versions prior to 5.0.1.

Column	Field	Description
1	Record Descriptor	Either DIAG_VR2TX_INTERIM_DESC (row containing column names for the DIAG_VR2TX_INTERIM record) or DIAG_VR2TX_INTERIM (a VR2Tx interim diagnostics data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Pings	Number of pings recorded in the previous 24 hours
9	Detections	Number of detections recorded in the previous 24 hours

3.4.6 HEALTH_VR2Tx

Health records contain receiver memory remaining and other health information logged daily at midnight.

Column	Field	Description
1	Record Descriptor	Either HEALTH_VR2TX_DESC (row containing column names for the HEALTH_VR2TX record) or HEALTH_VR2TX (a health event data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Memory Remaining (%)	Percentage of total memory capacity remaining
9	RTC Time	UTC date and time according to the receiver's real time clock. Used for troubleshooting purposes.

3.5 Configuration Records

3.5.1 CFG_CHANNEL

CFG_CHANNEL records contain information about the configuration of a receiver channel. Receivers with multiple channels (ex: the HR2 receiver) have multiple CFG_CHANNEL entries (one entry per channel).

Column	Field	Description
1	Record Descriptor	Either CFG_CHANNEL_DESC (row containing column names for the CFG_CHANNEL record) or CFG_CHANNEL (a channel configuration event data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Channel	Receiver channel number. Always Channel 1 for the VR2Tx.
9	Detection Type	Reports whether PPM or HR signal detected. Always PPM for the VR2Tx.
10	Frequency (kHz)	Channel frequency. Always 69 kHz for the VR2Tx.
11	Blanking (ms)	Unused for VR2Tx
12	Map ID	Code map used to decode detections
13	Coding ID	Unused for VR2Tx

3.5.2 CFG_STATION

CFG_STATION records contain information about the station (location) where the receiver is deployed.

Column	Field	Description
1	Record Descriptor	Either CFG_STATION_DESC (row containing column names for the CFG_STATION record) or CFG_STATION (a station configuration event data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Station Name	The name of the station (deployment location)
9	Latitude (degrees)	Station latitude
10	Longitude (degrees)	Station longitude

3.5.3 CFG_STUDY

CFG_STUDY records contain information about the study.

Column	Field	Description
1	Record Descriptor	Either CFG_STUDY_DESC (row containing column names for the CFG_STUDY record) or CFG_STUDY (a study configuration event data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Description	Description of the study

3.5.4 CFG_TRANSMITTER

CFG_TRANSMITTER records contain information about the settings of the receiver's self-transmitter. These records are generated at study initialization and when changes are made to the self-transmitter settings.

Column	Field	Description
1	Record Descriptor	Either CFG_TRANSMITTER_DESC (row containing column names for the CFG_TRANSMITTER record) or CFG_TRANSMITTER (a self-transmitter configuration event data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	Transmission Type	The coding system used to transmit. Always PPM for the VR2Tx.
9	Full ID	Full ID of the receiver's self-transmitter including code space (ex: A69-1601-62049)
10	ID	The ID of the receiver's self-transmitter without the code space (ex: 62049)
11	Power Level	The self-transmitter's transmission power level (ex: HIGH)
12	Min Delay (s)	The minimum time between transmissions of the self-transmitter
13	Max Delay (s)	The maximum time between transmissions of the self-transmitter

3.6 Other Records

These records are exported to CSV with all record type selections. They are related to clock events, data errors in the file, and identification of the data source file.

3.6.1 CLOCK_REF

CLOCK_REF records contain clock reference information. Clock reference information is date and time information from an external source such as your computer that is added to the receiver log file to help with correction for receiver clock drift. CLOCK_REF records are used in conjunction with CLOCK_SET records to calculate drift-corrected times ("Time" field) for the data in all record types.

Column	Field	Description
1	Record Descriptor	Either CLOCK_REF_DESC (row containing column names for the CLOCK_REF record) or CLOCK_REF (a clock reference data row)
2	Device Time (UTC)	UTC date and time of the data according to the receiver's clock (the uncorrected time).
3	Time	Date and time of the data following correction for receiver clock drift (the <i>corrected</i> time). The time offset of this time from UTC is indicated in the "Time Offset (h)" column.
4	Time Offset (h)	Offset from UTC time, in hours, of the value in the "Time" column. The time offset is as selected when the receiver log was exported to Fathom CSV format using the Fathom software. This column is blank if no time offset was applied at export.
5	Time Correction (s)	The difference between uncorrected ("Device Time (UTC)") and corrected ("Time") times.
6	Model	Receiver model that generated the data in this record. Always VR2Tx-69 for the VR2Tx.
7	Serial Number	Serial number of the receiver that generated the data in this record.
8	External Time (UTC)	UTC date and time of the external (reference) device (ex: UTC time calculated from your computer's time and time zone settings)
9	External Difference (s)	Difference between the external (reference) date and time ("External Time (UTC)") and the date and time according to the receiver's clock
10	Source	Source of the external date and time information (ex: OFFLOAD)

3.6.2 DATA_ERROR

DATA_ERROR records contain information about errors in the data file. They are used by Innovasea Support for troubleshooting purposes.

Column	Field	Description
1	Record Descriptor	Either DATA_ERROR_DESC (row containing column names for the DATA_ERROR record) or DATA_ERROR (a data error data row)
2	Type	The type of data error (RECORD or PAGE)
3	Error	The name of the error (ex: DECODER_EXCEPTION)
4	Page	Unused
5	Offset	Unused

3.6.3 DATA_SOURCE_FILE

DATA_SOURCE_FILE records contain information about the receiver log file that was used to generate the CSV output.

Column	Field	Description
1	Record Descriptor	Either DATA_SOURCE_FILE_DESC (row containing column names for the DATA_SOURCE_FILE record) or DATA_SOURCE_FILE (a data source file data row)
2	File Name	The name of the receiver log file that was used to create this CSV file
3	UUID	A unique identifier for the file. Used by Innovasea Support for troubleshooting purposes.
4	Type	File type information. Used by Innovasea Support for troubleshooting purposes.
5	Size	Size (in bytes) of the receiver log file that was used to create this CSV file
6	State	Indicates whether the receiver log file used to create this CSV file is an ORIGINAL or EDITED file